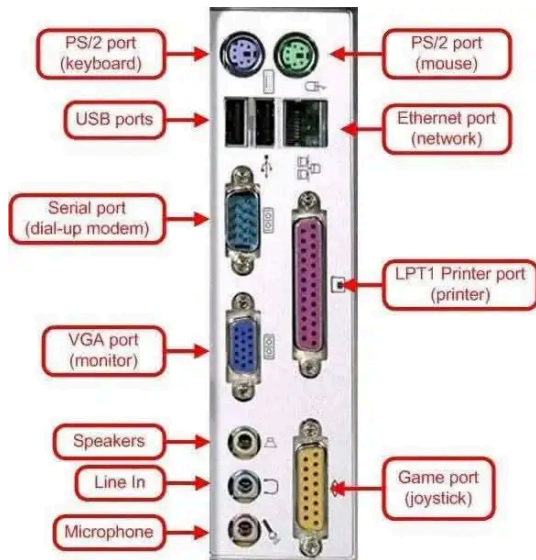


USB Part 1: Types and Versions

Why USB (Universal Serial Bus)?

Before USB, we had a mess of different ports and cables for each device:



- Serial ports for mice and modems
- Parallel ports for printers
- PS/2 ports for keyboards and mice
- SCSI for hard drives
- FireWire for high-speed devices

1. On top of that, each device had its own unique connector and communication method, leading to a nightmare of compatibility issues.
2. Also, power delivery was separate, so you needed different chargers for different devices.



The USB Solution

A unified system providing:

- **Power** → charges your phone
- **Data transfer** → copies files, streams video
- **Device discovery & drivers** → plug-and-play
- **Standard communication rules (protocols)** → everything speaks the same language

Real analogy: Like a universal power outlet + postal system + phone line, all in one cable!

Confusing USB Terms

1. **USB version** (1.0, 2.0, 3.0, etc.) → data transfer speed
2. **USB connector type** (Type-A, Type-B, Micro-USB, USB-C)
→ physical shape of the plug

USB protocol stack → the layered communication model that defines how data is structured and transmitted over USB (covered in Part 2)

USB power delivery (PD) → how much power it can provide and how it negotiates that power (covered in Part 3)

USB Versions (Data Transfer Speeds)

- **USB 1.0 (1996):** 1.5 Mbps (Low Speed) and 12 Mbps (Full Speed)
- **USB 2.0 (2000):** 480 Mbps (High Speed)
- **USB 3.0 (2008):** 5 Gbps (SuperSpeed)
- **USB 3.1 (2013):** 10 Gbps (SuperSpeed+)
- **USB 3.2 (2017):** 20 Gbps (SuperSpeed+)
- **USB4 (2019):** 40 Gbps (Thunderbolt 3 compatible)

USB Versions				Sign	Transfer Rate	Theoretical Speed
Current Version Name		Market Code	Original Name			
USB 2.0	LowSpeed	Low Speed	USB 1.0		1.5Mbps	0.1875MB/s
	FullSpeed	Full Speed	USB 1.1		12Mbps	1.5MB/s
	HiSpeed	Hi-Speed	USB 2.0		480Mbps	60MB/s
USB 3.2	Gen 1	SuperSpeed USB	USB 3.1GEN1		5Gbps	625MB/s
	Gen 2	SuperSpeed USB 10Gbps	USB 3.1GEN2		10Gbps	1250 MB/s
	Gen 2x2	SuperSpeed USB 20Gbps	N/A		20Gbps	2500MB/s
USB 4					40Gbps	5GB/s

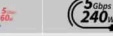
Notice that the theoretical maximum speeds are often not achieved in real-world use due to various factors like cable quality, device capabilities, and protocol overhead.

Notice how the speed increases with each version, but the connector type doesn't necessarily change.

Device port USB logos

Logo Name	Packaging Logo	Port Logo	Device Power Port Logo
USB 40Gbps			
USB 20Gbps			
USB 10Gbps			
USB 5Gbps			

Cable USB logos

Logo Name	Packaging Logo	Cable Logo
USB 40Gbps Cable	 	 
USB 20Gbps Cable	 	 
USB 5Gbps Cable	 	 
Hi-Speed USB Cable	 	 

Charger USB logos

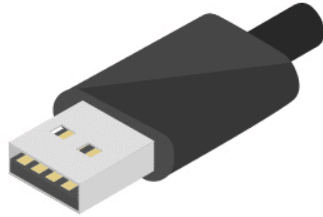
Logo Name	Packaging Logo
Certified USB Charger	
Certified USB Fast Charger	

- You can have a USB 2.0 Type-A/B connector.
- You can have a USB 3.0 Type-A/B connector.
- You can have a USB 3.0 Type-C connector.

USB Connectors

The physical connectors come in various shapes and sizes, but they all follow the same underlying communication protocols:

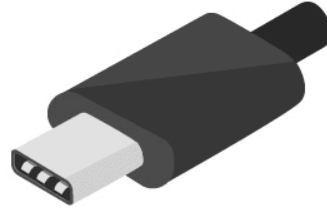
Type	Description	Common Usage
Type-A	Classic rectangular plug	Computers, chargers
Type-B	Square-shaped connector	Printers, external drives
Mini-USB	Smaller than Type-B	Older cameras, devices
Micro-USB	Thinner, compact	Older smartphones, peripherals
USB-C	Reversible, high-speed + power	Modern phones, laptops



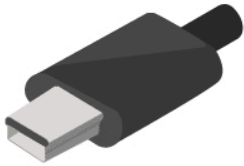
USB TYPE A



USB TYPE B



USB TYPE C



USB MINI A



USB MINI B



USB MICRO C



USB MICRO C



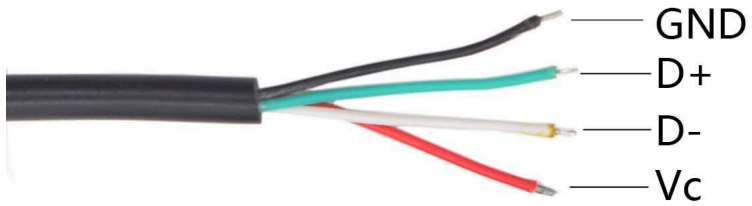
USB MICRO B SUPER SPEED

USB Type A and Type B

- USB Type A is the standard rectangular connector that plugs into **computers** and chargers.
- USB Type B is a square-shaped connector typically used for **external drives** such as printers.



USB Type A 4 Wires



- Vcc (Red) → +5V power
- D- (White) → Data -
- D+ (Green) → Data +
- GND (Black) → Ground

USB Micro and Mini (Type B variants)

For smaller devices, USB Type B connectors were too large, so new variants were developed:



- For old IoT devices, you might still find Micro-USB type B.
- Mini-USB type B was widely used for older smartphones and peripherals such as digital cameras.

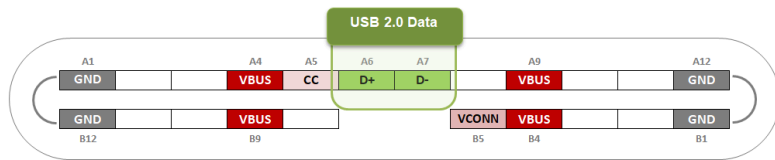
Even mini-USB type A was invented.



- Remember that the connector type A is for hosts (like computers), while type B is for devices (like printers).
- Nobody wanted to use mini-USB type A, so it never became popular.

USB Type C

USB Type C was introduced in 2014 as a new standard for USB connectors.



- It supports USB 2.0, USB 3.0, and USB PD, making it a versatile connector for a wide range of devices.
- It replaces the older USB Type A and Type B connectors with a smaller, reversible design that can be plugged in either way.
- Be careful some USB-C cables only support USB 2.0 with fewer wires.

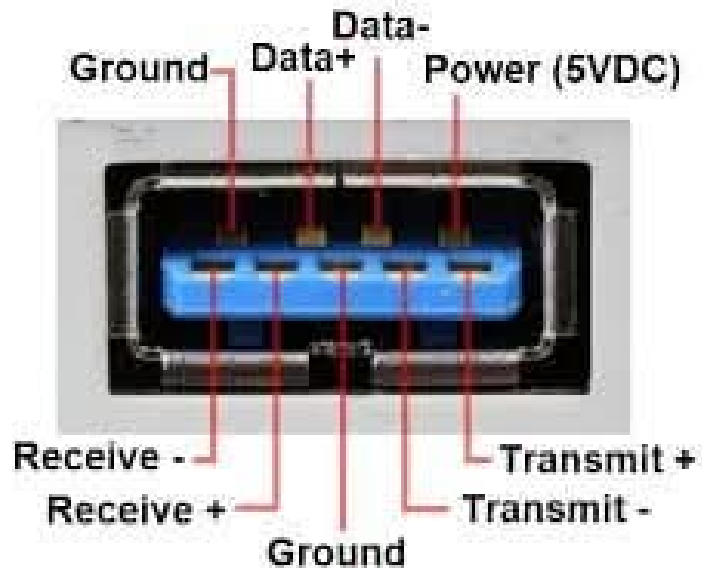
USB 3.0

USB 3.0 was introduced in 2008.

- For USB Type C that was invented in 2014, so it was made for USB 3.0.
- But older types such as, USB A/B, has only four wires that were not enough to support the higher data transfer speeds.

What's the solution?

New Wires for USB 3.0



- USB 3.0 introduced a new connector design for Type A and Type B.
- It added five additional wires to support the SuperSpeed data transfer mode, allowing for speeds up to 5 Gbps.

Notice the blue color inside the connector, which is a common visual indicator of USB 3.0 ports.

USB type A Pin Out



USB version 2.0 (4 pins)



Pin Out	
1	(+) 5V
2	Data-
3	Data+
4	(-) Gnd



Male Plug

Female Receptacle

USB version 3.0 (9 pins)



Pin Out	
5	Rx-
6	Rx+
7	Gnd
8	Tx-
9	Tx+



USB version 3.0 is pin compatible to USB version 2.0

Notice the two more pairs (total 3 pairs) of wires in the USB 3.0 Type A connector compared to the USB 2.0 version.

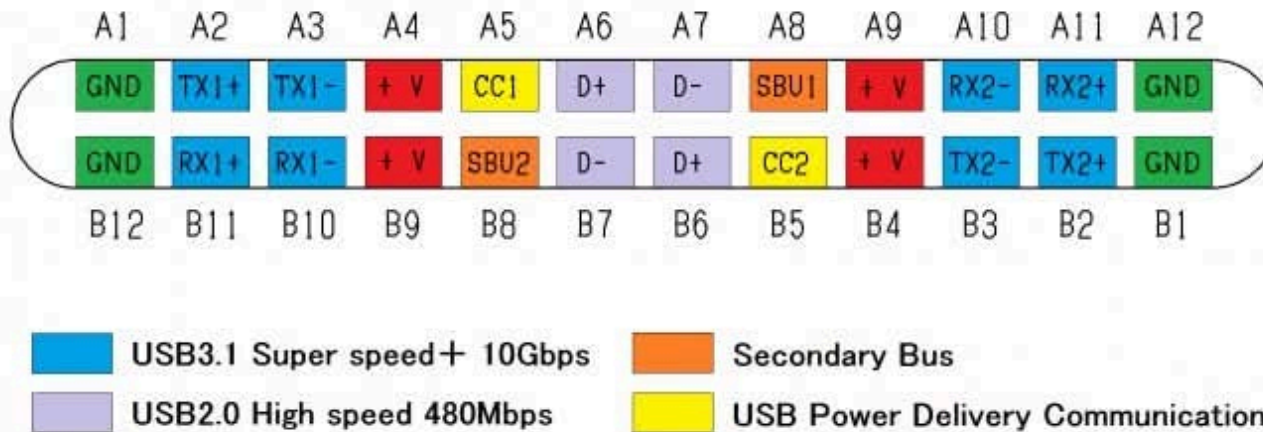
USB Type B 3.0 Connectors

Because of the additional 5 wires, USB 3.0 Type B connectors are larger than their USB 2.0 counterparts, which can be an issue for smaller devices.



USB Type C 3.0 Connectors

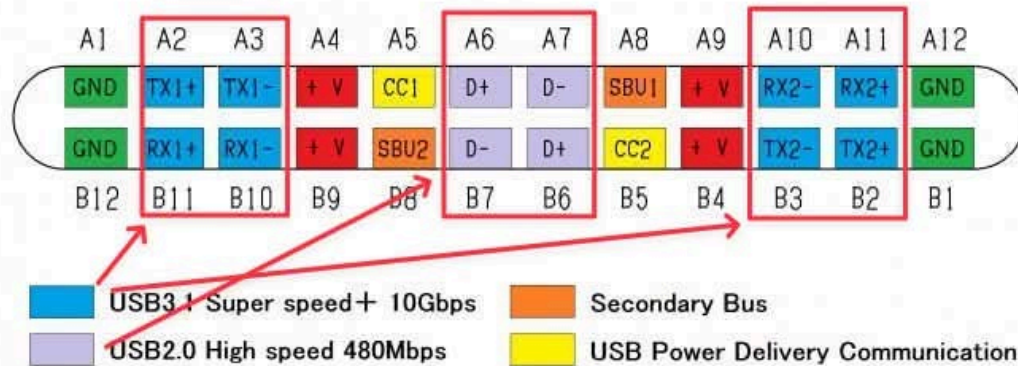
USB Type-C Connector Pin Assign



- USB Type C connectors have 24 pins, which allow for higher data transfer speeds and power delivery capabilities compared to older USB types.
- Pins are arranged in a symmetrical matter to be reversible.

Communication Pins in USB Type C

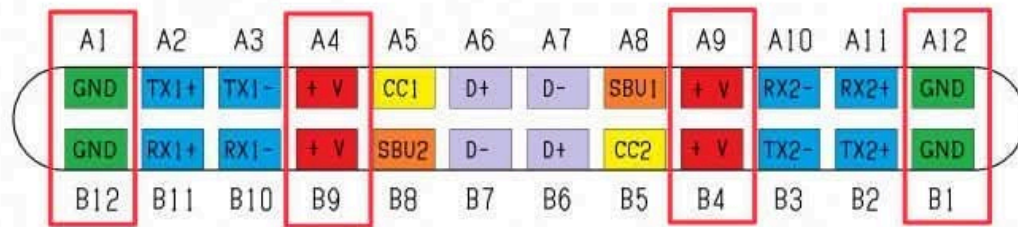
USB Type-C Connector Pin Assign



- USB Type C has one pair of wires for USB 2.0 data transfer (D+ and D-).
- USB Type C connector two pairs of wires for high speed (TX+ and TX-/RX+ and RX-).

Power pins in USB Type C

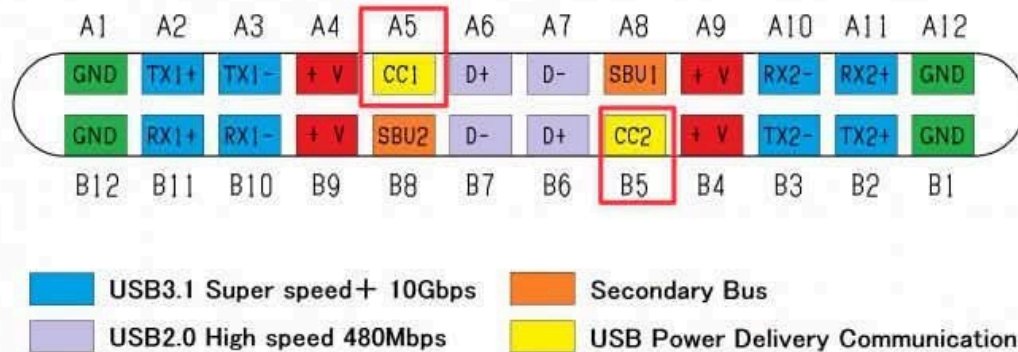
USB Type-C Connector Pin Assign



- USB Type C has two pairs of Power (VBUS, +V) and Grounds (GND) to provide up to 5A.
- It allows for higher power delivery (up to 100W with USB PD).

Configuration Channel (CC) Pins in USB Type C

USB Type-C Connector Pin Assign

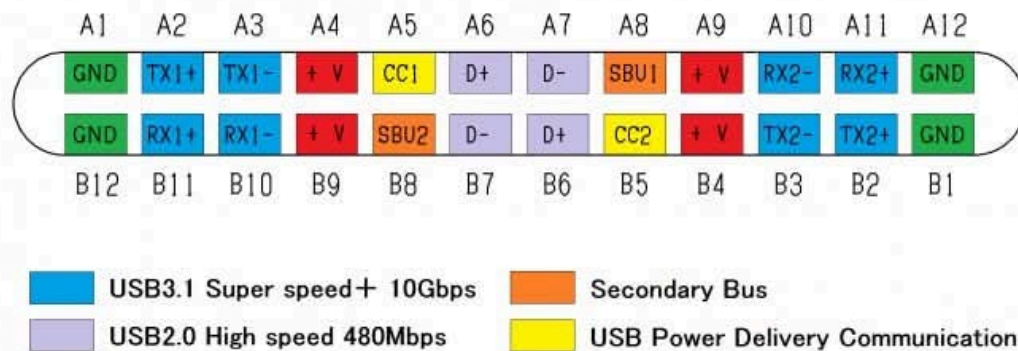


- USB Type-C includes **two CC pins (CC1, CC2)**.
- They are used to detect:
 - **Cable orientation** (plug flip detection)
 - **Device roles** (Source vs Sink)

- CC pins advertise **default current levels**:
 - 500 mA / 900 mA (USB default)
 - 1.5 A
 - 3 A
- They carry **USB Power Delivery (PD) communication**:
 - Negotiates voltage (5 V → 9 V → 15 V → 20 V → higher with EPR)

SBU (Sideband Use) pins in USB Type C

USB Type-C Connector Pin Assign



- USB Type-C provides two SBU pins (SBU1, SBU2).

- They are **auxiliary signal lines**, not used for standard USB data.
- Primarily used in **Alternate Modes**, such as:
 - DisplayPort Alt Mode (AUX channel)
 - Analog Audio Adapter Mode
 - Vendor-specific extensions
- SBU pins remain **idle in normal USB operation**.

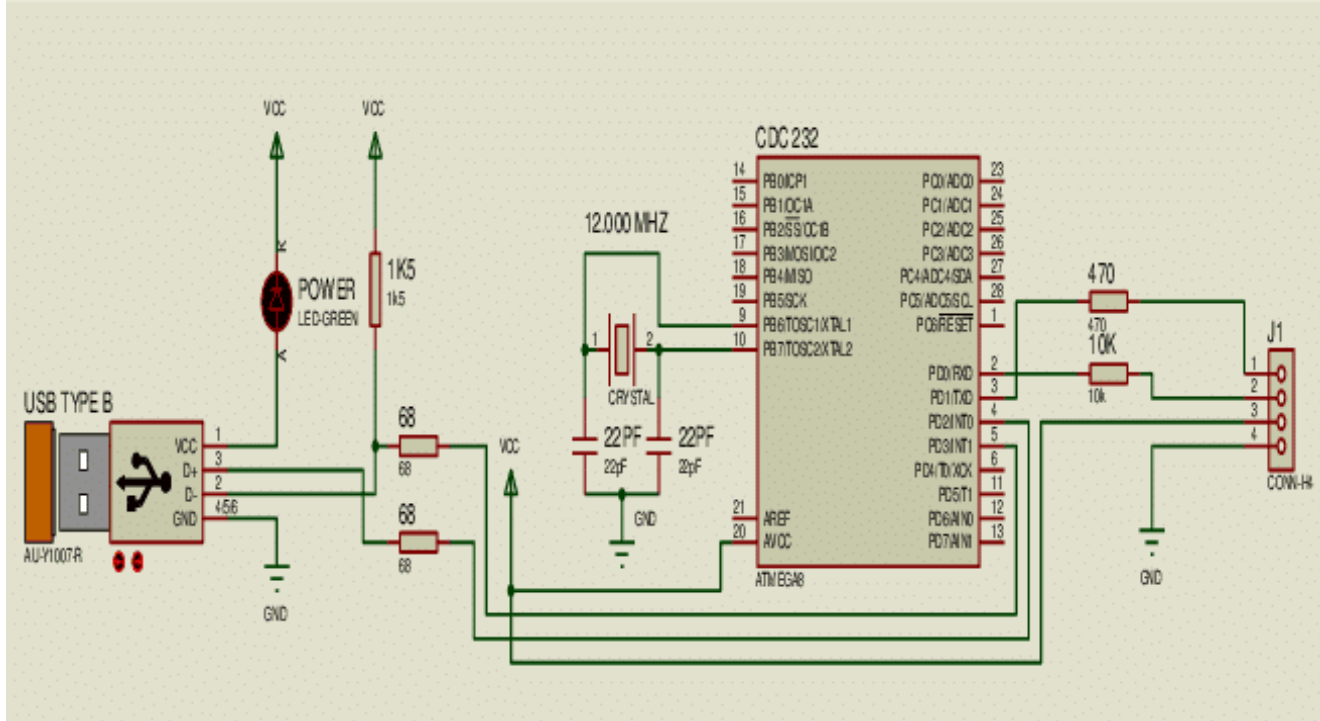
USB and IoT

In most IoT applications, USB is used for:

1. Communication between computers and IoT devices/boards.



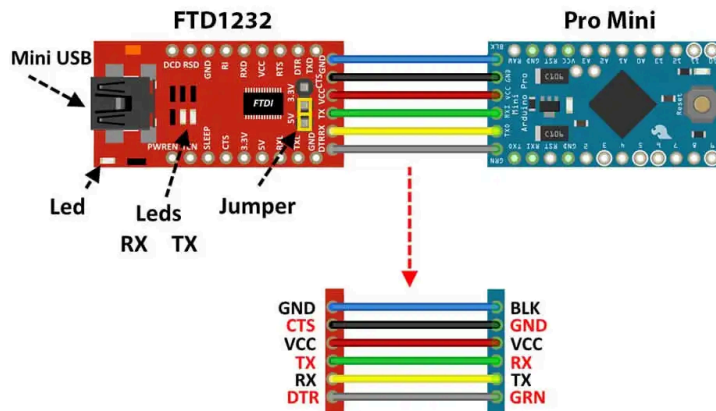
- For example, Arduino nano has a USB port and a microcontroller that processes data from the PC as if it is a serial port.
- The PC should install the appropriate drivers to recognize the Arduino as a serial device, allowing for easy communication and programming.



1. The USB port is for communication and power.
2. This dedicated microcontroller is used only for USB communication.

Nano to Pro Mini

1. When we deploy IoT devices, we often don't need the USB communication anymore.
2. So we can remove the USB port and use a smaller board like the Pro Mini, which is cheaper and more compact.
3. When we need to program the Pro Mini, we can use a dongle such as FTD1232.



Powering IoT devices

- For example, many IoT devices can be powered through a USB port (5V), which provides a convenient and standardized way to supply power.
- In some USB cables, there are only two power wires (Vcc and GND) without data wires, which are used solely for charging or powering devices.
- When we need more power ($>5V$), we can use USB PD with a USB-C connector, which allows for higher power delivery and faster charging.

Summary

Now, you can understand all USB versions and types.

